

Interview-Oriented Business Case & Formula Questions

These questions are designed around the **Enterprise Lookup & Reference Functions dataset** created for Lecture 05 and cover exactly the concepts taught:

Cell Referencing → Absolute Referencing → VLOOKUP → HLOOKUP → XLOOKUP → MATCH → INDEX → INDEX + MATCH → Two-Way Lookup

I've divided them into **theoretical interview questions, formula-based questions, and real business case scenarios**, progressing from foundational to interview-level analytical problems.

Section A — Theoretical & Conceptual Interview Questions

Q1. Relative vs Absolute Cell Reference

Interview Question: What is the difference between a relative cell reference and an absolute cell reference in Excel? Give one business scenario where each would be appropriate.

Expected concepts:

- A2 changes when copied.
- \$A\$2 remains fixed.
- Relative reference: row-wise sales calculation.
- Absolute reference: fixed tax, commission, exchange rate, or lookup-table range.

Hint: Think about what should change and what should remain constant when dragging a formula down.

Q2. Why Absolute References Matter in VLOOKUP

A formula is written as:

=VLOOKUP(D2,Product_Master!A2:J81,2,FALSE)

It is then copied down 1,600 transaction rows.

Question: What problem can occur, and how would you correct it?

Hint: Observe what happens to A2:J81 when the formula moves down.

Q3. VLOOKUP vs XLOOKUP

Interview Question: Your manager asks why you prefer XLOOKUP over VLOOKUP in a modern Excel reporting solution. Give at least four technical reasons.

Expected concepts:

- Exact match by default



- Left and right lookup
- No hard-coded column index
- Built-in if_not_found
- Search direction control
- Multiple-column return

Q4. Why Can VLOOKUP Not Directly Perform a Left Lookup?

The Product_Master sheet has:

Product ID | Product Name | Category | Brand | Unit Price

Suppose you know the **Product Name** and want to return the **Product ID**.

Question: Why can't a standard VLOOKUP directly solve this using the current table structure? Which functions would you use instead?

Hint: Consider the position of the lookup column relative to the return column.

Q5. Exact Match vs Approximate Match

Explain the difference between:

=VLOOKUP(A2,\$F\$2:\$G\$10,2,FALSE)

and:

=VLOOKUP(A2,\$F\$2:\$G\$10,2,TRUE)

Give one valid business use case for each.

Expected answer direction:

- FALSE: Product ID, Customer ID, Employee ID.
- TRUE: Commission slabs, tax brackets, rating bands, discount tiers.

Q6. INDEX vs MATCH

Interview Question: What does INDEX do, what does MATCH do, and why are they commonly combined?

Hint:

MATCH → Finds where

INDEX → Returns what



Q7. Duplicate Lookup Values

A Product Master accidentally contains the same Product ID twice.

Question: What would standard VLOOKUP, default XLOOKUP, and INDEX + MATCH normally return?

Follow-up: Why can duplicate lookup keys be dangerous in business reporting?

Q8. Understanding the #N/A Error

An analyst applies this formula:

=XLOOKUP(D2,Product_Master!\$A\$2:\$A\$81,Product_Master!\$B\$2:\$B\$81)

Some transactions return #N/A.

Question: List at least four possible business or data-quality reasons.

Hint: Consider missing records, extra spaces, data types, and incorrect IDs.

Q9. HLOOKUP vs VLOOKUP

Interview Question: What determines whether you should use VLOOKUP or HLOOKUP?

Your monthly sales targets are arranged as:

	Jan	Feb	Mar	Apr
EMP-001	500K	520K	550K	580K
EMP-002	450K	480K	510K	540K

Would HLOOKUP alone be sufficient to dynamically retrieve a target using **both Employee ID and Month**? Explain.

Q10. Why Is XLOOKUP More Structurally Robust?

Consider:

=VLOOKUP(D2,\$A\$2:\$J\$81,6,FALSE)

versus:

=XLOOKUP(D2,\$A\$2:\$A\$81,\$F\$2:\$F\$81)

Question: What happens if a new column is inserted between existing columns? Why is XLOOKUP generally safer?

Section B — Formula-Based Business Questions

Q11. Retrieve Product Name Using VLOOKUP

In Sales_Transactions, D2 contains the Product ID.

Business Requirement: Retrieve the corresponding Product Name from Product_Master.

Write a VLOOKUP formula that:

- Uses exact matching.
- Uses proper absolute references for the lookup table.
- Can safely be copied down.

Hint: Product Name is the second column of Product_Master.

Q12. Retrieve Unit Price Using VLOOKUP

The finance team needs the standard Unit Price for every transaction.

Task: Using the Product ID in D2, retrieve the Unit Price from Product_Master.

Constraint: Use VLOOKUP.

Hint: Unit Price is column 6 within the Product Master table.

Q13. Retrieve Customer Region Using XLOOKUP

The marketing team wants to perform regional sales analysis.

Task: Using the Customer ID in C2, return the customer's Region from Customer_Master.

If no customer is found, display:

Customer Not Found

Constraint: Use XLOOKUP.

Q14. Retrieve Employee Manager

The sales director wants every transaction mapped to the responsible salesperson's manager.

Task: Use the Salesperson ID in E2 to retrieve the corresponding Manager Name from Employee_Master.

Constraint: Use XLOOKUP.

Business question: How would this help in management-level sales performance reporting?

Q15. Perform a Left Lookup

Suppose cell A2 contains:

Wireless Mouse Pro

Task: Return its corresponding Product ID from Product_Master.

Constraint: Use XLOOKUP.

Interview follow-up: Why is XLOOKUP more suitable than standard VLOOKUP here?

Q16. Find the Position of a Product Using MATCH

Using the Product ID in D2, find its relative position within:

Product_Master!\$A\$2:\$A\$81

Constraint: Use an exact MATCH.

Follow-up: If the formula returns 45, what does that actually mean?

Q17. Retrieve Product Name Using INDEX Only

Assume you already know that the required product exists at position 25 within:

Product_Master!\$B\$2:\$B\$81

Task: Write an INDEX formula to return that Product Name.

Follow-up: Why is hard-coding the row position not suitable for dynamic business reporting?

Q18. Retrieve Cost Price Using INDEX + MATCH

The finance team needs Cost Price for margin analysis.

Task: Using Product ID from D2, return the corresponding Cost Price from Product_Master.

Constraint: Use:

INDEX + MATCH

Hint:

- MATCH searches the Product ID column.
 - INDEX returns from the Cost Price column.
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Q19. Retrieve Customer Loyalty Tier

The CRM team needs to classify transactions based on customer loyalty.

Task: Using Customer ID from C2, retrieve the customer's Loyalty Tier.

Possible values include:

Bronze

Silver

Gold

Platinum



Constraint: Use INDEX + MATCH.

Q20. Retrieve Multiple Product Attributes with One XLOOKUP

The operations team wants to retrieve all of the following using a single Product ID:

- Product Name
- Category
- Sub-Category
- Brand
- Unit Price

Task: Write one XLOOKUP formula capable of returning all five attributes horizontally.

Hint: Your return array should span multiple adjacent columns.

Section C — Advanced Business Case Scenarios

Q21. Calculate Gross Sales Using XLOOKUP

The Sales_Transactions sheet contains:

- D2 = Product ID
- F2 = Quantity

The Unit Price is stored separately in Product_Master.

Business Requirement: Calculate:

Gross Sales = Quantity × Unit Price

Write one formula that directly retrieves Unit Price using XLOOKUP and calculates Gross Sales.

Follow-up: Why might a business prefer retrieving prices from a controlled Product Master rather than manually storing them in every transaction row?

Q22. Calculate Net Sales After Discount

The transaction sheet contains:

- D2 = Product ID
- F2 = Quantity
- G2 = Discount %

Business Requirement:

Gross Sales = Quantity × Unit Price



Net Sales = Gross Sales $\times$ (1 - Discount %)

Write a single Excel formula using XLOOKUP.

Q23. Classify High-Value Transactions

Management defines an order as:

High Value $\rightarrow$ Gross Sales $>$ ₹100,000

Regular $\rightarrow$ Gross Sales $\leq$ ₹100,000

Task: Write one formula combining:

IF + XLOOKUP

to classify every transaction.

Q24. Identify Products Below Reorder Level

The Product_Master contains:

- Stock Quantity
- Reorder Level

The procurement team needs to flag products as:

Reorder Required

Sufficient Stock

Task: Write a logical formula to compare current Stock Quantity against Reorder Level.

Interview follow-up: Is a lookup actually necessary if both values already exist in the same row of Product_Master? Explain.

This is a trap question. A good analyst should not use a lookup when a simple direct comparison is sufficient.

Q25. Find Employee Tenure Using XLOOKUP + DATEDIF

The HR analytics team wants to calculate each salesperson's completed years of service.

The employee's Joining Date exists in Employee_Master.

Task: Using Salesperson ID from E2:

1. Retrieve Joining Date with XLOOKUP.
2. Calculate completed years until today.
3. Use DATEDIF and TODAY.

Required functions:

XLOOKUP + DATEDIF + TODAY

Q26. Two-Way Employee Target Lookup

The Monthly_Targets sheet contains:

- Employee IDs vertically.
- Months horizontally.
- Sales targets at their intersections.

Suppose:

A2 = EMP-005

B2 = Mar

Business Requirement: Dynamically retrieve the March target for EMP-005.

Constraint: Use:

INDEX + MATCH + MATCH

Interview follow-up: Explain the role of each MATCH function.

Q27. Compare Actual Sales Against Monthly Target

Management wants to classify employee performance as:

Target Achieved

Target Not Achieved

You have:

- Employee ID
- Selected Month
- Actual Sales
- Monthly targets stored in Monthly_Targets

Task: Design a formula combining:

IF + INDEX + MATCH + MATCH

to compare actual sales against the dynamically retrieved target.

Hint:

IF(Actual Sales >= Retrieved Target, "Target Achieved", "Target Not Achieved")



Q28. Handle Missing Product IDs Professionally

Some deliberately unmatched Product IDs exist in the dataset.

Management does not want reports displaying:

#N/A

Task: Write three different solutions:

1. Using XLOOKUP's built-in [if_not_found] argument.
2. Using IFNA.
3. Using IFERROR.

Interview follow-up: Why is IFNA sometimes more precise than IFERROR for lookup operations?

Q29. Retrieve the Last Matching Record Using XLOOKUP

Suppose a dataset contains multiple transactions for the same Customer ID.

Management wants the **last matching transaction** rather than the first.

Question: How can XLOOKUP's search_mode argument be used to search from bottom to top?

Challenge: Write the conceptual formula using:

search_mode = -1

Interview relevance: This is a common advanced XLOOKUP question because many candidates know only the first three arguments.

Q30. Dynamic Product Price & Discount Analysis Using Lookup Functions

A company maintains product pricing information separately in the Product_Master sheet, while individual orders are recorded in Sales_Transactions.

The transaction sheet contains:

Product ID

Quantity

Discount %

The Product Master contains:

Product ID

Product Name

Category

Unit Price

Cost Price

Business Requirement

For each transaction, management wants you to retrieve the following information:

- Product Name
- Product Category
- Unit Price

Then calculate:

Gross Sales = Quantity × Unit Price

and:

Net Sales = Gross Sales × (1 - Discount %)

Finally, classify the transaction as:

High Value → Net Sales > ₹100,000

Medium Value → Net Sales > ₹50,000

Regular → Net Sales ≤ ₹50,000

Tasks

1. Retrieve the **Product Name** using VLOOKUP.
2. Retrieve the **Product Category** using XLOOKUP.
3. Retrieve the **Unit Price** using INDEX + MATCH.
4. Calculate **Gross Sales**.
5. Calculate **Net Sales after Discount**.
6. Use a logical function to classify each transaction as High Value, Medium Value, or Regular.
7. Explain where **relative cell references** should be used in these formulas.
8. Explain where **absolute cell references** should be used and why.

Interview Follow-Up Questions

- a. Why would you use absolute references for the Product Master lookup range?
- b. What would happen if you copied a VLOOKUP formula down 1,600 rows without locking the table array?
- c. Which of the three lookup approaches—VLOOKUP, XLOOKUP, or INDEX + MATCH—would you choose for a new Excel project, and why?
- d. If a new column is inserted into the Product Master, which lookup approach is most likely to be affected?
- e. Why is it generally better to maintain product pricing in a centralized Product Master rather than manually entering the price in every transaction?

